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SIMATS ENGINEERING

CO1 - ASSESSMENT TOOL

2 ERROR ANALYSIS TASK

COURSE NAME: Deep Learning
FACULTY : Dr.Arul Raj A.M

COURSE CODE: MLA0403

COURSE OUTCOME COVERED

CO 1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BTL-4)

CO1 Weightage: 15%

Assessment Tool Weightage: 30%

Duration: 30 minutes

Marks: 25

Code of Conduct:

I P.Mokshith Reddy / 192425149 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

P.mokshith reddy

QUESTIONS:**Total Marks:25****1. Error Analysis in Maximum Likelihood Estimation (MLE)**

A student builds a binary classifier using Maximum Likelihood Estimation. The likelihood function is incorrectly written as the sum of probabilities instead of the product. The model produces poor parameter estimates even with large data.

Tasks:

a) Identify the conceptual error in the likelihood formulation.

Answer :

The conceptual error is that the likelihood function is written as the sum of probabilities instead of the product of probabilities. In MLE, each observation is assumed to be independent, so the joint likelihood is obtained by multiplying the probabilities of all observations.

b) Explain how this mistake affects parameter estimation.

Answer :

Using the sum instead of the product does not represent the joint probability of the dataset. As a result, the optimization process estimates incorrect parameters that do not maximize the true likelihood. Even with large datasets, the model learns inaccurate parameter values, leading to poor prediction performance.

c) Suggest the correct mathematical formulation and reasoning.

Answer :

The correct likelihood function is :

$$L(\theta) = \prod_{i=1}^n P(x_i | \theta)$$

where:

- x_i = observations
- θ = model parameters

The product is used because all observations are assumed to be independent. Maximizing this likelihood provides the parameter values that best explain the observed data.

d) How would using log-likelihood fix computational issues?

Answer :

The log-likelihood is

$$\log L(\theta) = \sum_{i=1}^n \log P(x_i | \theta)$$

Taking the logarithm converts the product into a sum, making computations easier. It also prevents numerical underflow caused by multiplying many small probabilities and simplifies differentiation during optimization.

2. Error Diagnosis in Perceptron Learning

While implementing a Perceptron, a student observes that the model fails to converge even after many iterations on a linearly separable dataset.

Tasks:

a) List possible implementation errors in the learning algorithm.

Answer :

Possible implementation errors include:

- Incorrect weight update equation.
- Missing bias update.
- Wrong prediction function.
- Incorrect learning rate.
- Errors in data preprocessing.
- Stopping training too early.
- Incorrect class labels.

b) Analyze how incorrect learning rate or weight updates affect convergence.

Answer :

If the learning rate is too large, weight updates become unstable and may overshoot the optimal solution. If it is too small, convergence becomes very slow. Incorrect weight update equations prevent the perceptron from learning the correct decision boundary, causing failure to converge.

c) Identify dataset-related issues that may cause this problem.

Answer :

Possible dataset issues include:

- Incorrect data labels.
- Duplicate or noisy samples.
- Features with different scales.
- Presence of outliers.
- Dataset not actually linearly separable.

d) Propose modifications to ensure convergence.

Answer :

To ensure convergence:

- Normalize the input features.
- Use the correct perceptron update rule.
- Select an appropriate learning rate.
- Update bias correctly.
- Remove noisy or mislabeled data.
- Verify that the dataset is linearly separable.

3. Backpropagation Gradient Error Analysis

A Multilayer Perceptron trained using Backpropagation shows increasing loss during training instead of decreasing.

Tasks:

a) Identify possible mathematical or coding errors in gradient computation.

Answer :

Possible errors include:

- Incorrect derivative calculation.
- Wrong chain rule implementation.
- Incorrect weight update direction.
- Missing activation derivative.
- Programming errors in matrix multiplication.
- Incorrect learning rate implementation.

b) Explain the role of activation functions in gradient flow.

Answer :

Activation functions introduce non-linearity into neural networks. During backpropagation, their derivatives determine how gradients flow through the network. Suitable activation functions help maintain stable gradients, while poor choices may reduce gradient values and slow learning.

c) Analyze how vanishing or exploding gradients affect training.

Answer :

Vanishing gradients occur when gradients become extremely small, preventing early layers from learning effectively. Exploding gradients occur when gradients become excessively large, causing unstable weight updates and increasing the training loss.

d) Suggest techniques to fix the issue (e.g., normalization, initialization).

Answer :

Techniques include:

- Xavier or He weight initialization.
- Batch Normalization.
- Gradient Clipping.
- ReLU activation instead of sigmoid.
- Proper learning rate selection.
- Adaptive optimizers such as Adam.

4. Gradient Descent vs SGD Error Investigation

A model trained using Gradient Descent converges very slowly, while the same model with Stochastic Gradient Descent shows unstable loss fluctuations.

Tasks:

a) Explain why batch gradient descent is slow in large datasets.

Answer :

Batch Gradient Descent computes gradients using the entire training dataset before updating the weights. For large datasets, this requires significant computation in each iteration, making training slow and memory-intensive.

b) Analyze the cause of fluctuations in SGD.

Answer :

Stochastic Gradient Descent updates weights using one training example at a time. Since each sample provides a different gradient estimate, the optimization path becomes noisy, resulting in fluctuations in the loss function.

c) Compare the error behavior of Mini-Batch Gradient Descent.

Answer :

Mini-Batch Gradient Descent updates weights using a small group of samples. It reduces the noise of SGD while remaining much faster than Batch Gradient Descent. This provides more stable convergence with efficient computation.

d) Recommend the best approach and justify with error trade-offs.

Answer :

Mini-Batch Gradient Descent is generally the best approach because it balances computational efficiency and convergence stability. It reduces the slow computation of Batch Gradient Descent while minimizing the large fluctuations observed in SGD, resulting in faster and more reliable learning.

5. Curse of Dimensionality Error Analysis

A machine learning model built using high-dimensional data performs well on training data but poorly on test data due to overfitting, related to the Curse of Dimensionality.

Tasks:

a) Explain how high dimensionality increases model error.

Answer :

As the number of features increases, the feature space becomes sparse. The model requires significantly more training data to learn meaningful patterns. With limited data, it memorizes noise instead of learning general relationships, increasing prediction error on unseen data.

b) Identify signs of overfitting in this scenario.

Answer :

Signs of overfitting include:

- Very high training accuracy.
- Low testing accuracy.
- Large gap between training and testing performance.
- Poor generalization to new data.
- High model variance.

c) Analyze why distance-based learning fails in high dimensions.

Answer :

In high-dimensional spaces, the distances between data points become nearly identical. As a result, algorithms such as K-Nearest Neighbors cannot effectively distinguish between nearby and distant samples, reducing classification accuracy.

d) Suggest dimensionality reduction or regularization techniques to improve performance.

Answer :

Recommended techniques include:

- **Principal Component Analysis (PCA):** Reduces feature dimensions while preserving important information.
- **Feature Selection:** Removes irrelevant and redundant features.

- **L1 (Lasso) Regularization:** Eliminates less important features by shrinking coefficients to zero.
- **L2 (Ridge) Regularization:** Prevents overfitting by penalizing large weights.
- **Dropout (for Neural Networks):** Randomly disables neurons during training to improve generalization.

These techniques reduce model complexity, improve learning efficiency, and enhance performance on unseen data.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand